

Multi-Armed Bandits-Based Optimization of Decision Trees

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Abstract—Decision trees, without appropriate constraints, can easily become overly complex and prone to overfit, capturing noise rather than generalizable patterns. To resolve this problem, pruning operation is a crucial part in optimizing decision trees, as it not only reduces the complexity of trees but also decreases the probability of generating overfit models. The conventional pruning techniques like Cost-Complexity Pruning (CCP) and Reduced Error Pruning (REP) are mostly based on greedy approaches that focus on immediate gains in performance while pruning nodes of the decision tree. However, this might result in a lower generalization in the long run, compromising the robust ability of the tree model when introduced to unseen data samples, particularly when trained with small and complex datasets. To address this challenge, we are proposing a Multi-Armed Bandits (MAB)-based pruning approach, a reinforcement learning (RL)-based technique, that will dynamically prune the tree to generate optimal decision tree with better generalization. Our proposed approach assumes the pruning process as an exploration-exploitation problem, where we are utilizing the MAB algorithms to find optimal branch nodes to prune based on feedback from each pruning actions. Experimental evaluation on several benchmark datasets, demonstrated that our proposed approach results in better predictive performance compared to the traditional ones. This suggests the potential of utilizing MAB for a dynamic and probabilistic way of decision tree pruning, in turn optimizing decision tree-based model.

Index Terms—Decision Tree Pruning, Optimization, Multi-Armed Bandits, Reinforcement Learning.

I. INTRODUCTION

According to few studies [1] and [2], one major drawback of most algorithms utilized in machine learning (ML) is their lack of transparency. As machine learning is being applied in all sectors to make things more efficient in recent times, users prefer models that offer explanations about their decision making process. Being a transparent and interpretable algorithm, [3], along with its ease of implementation and low computational cost, decision tree based models are becoming more popular day by day in the sector of interpretable machine learning applications [4]. But, a key limitation of decision tree is its tendency to overfit. This problem of overfitting mainly occurs when the tree becomes too complex and starts to capture noises instead of capturing generalizable patterns in the datasets [5].

Pruning decision tree is an innovative way of mitigating the effect of overfitting and improving model's generalization ability [6]. By strategically removing branches that offer minimal

contribution to model's predictive performance, pruning not only optimize the decision tree but also increase its robustness. There are few existing methods for decision tree pruning, such as Reduced Error Pruning and Cost-Complexity Pruning. REP reduces decision tree complexity by removing branches iteratively based on its performance on a validation set, while CCP uses a penalty term to prune the tree while balancing model's complexity and accuracy. Although effective, they rely heavily on heuristic-based criteria or computationally expensive cross-validation procedures. These characteristics limit their scalability, particularly for real world applications.

As pruning a decision tree requires finding best branch node to prune at that moment without hampering the model's predictive performance, it can be designed as a sequential decision making problem. Methods like reinforcement learning shows great potential in this type of decision making tasks. Studies [7] and [8] have proven its effectiveness in optimizing sequential decision making tasks. Among various RL framework, Multi-Armed Bandits is really simple and easy to use. It offers sequential decision making in a dynamic way based on iterative feedback, focusing on balancing exploration and exploitation. Unlike current methods for pruning that heavily depend on stiff rules, MAB-based approach can learn an optimal pruning strategy by using feedback from pruning different tree branches to identify the best action.

Furthermore, it could also offer a probabilistic decision making, which can further improve the pruning operation. These characteristics of MAB make it an adaptive decision making framework, which is capable of delivering an efficient and data-driven pruning technique, optimizing decision tree.

In this paper, we propose a Multi-Armed Bandits-based decision tree pruning framework that dynamically selects branch nodes of decision tree for pruning with an objective of improving model's ability of generalization, interpretability and predictive performance. Our approach formulates the pruning operation as an exploration-exploitation problem where each branch node of decision tree is treated as an arm in a bandit framework and pulling an arm represents pruning a branch node. This approach iteratively updates its pruning action based on feedback from each actions. We evaluate our method on few benchmark datasets and compare its performance against few conventional pruning techniques.

The remainder of this paper is organized as follows: Section II reviews related work in decision tree pruning, reinforcement learning and multi-armed bandits in optimization and decision trees. Section III outlines the proposed MAB-based pruning approach, while Section IV details its evaluation and results. Section V discusses study limitations and Section VI concludes with future research directions.

II. RELATED WORK

Pruning operation is important for any network type machine learning algorithms, such as neural network, decision tree etc. It makes the network-based models more optimized while offering better predictive performance. Recently, RL-based methods showed great potential in this sector as they can adapt based on their past actions and strategize for future decisions. In this section, we will review the literature on decision tree pruning, reinforcement learning in optimization and the integration of MAB algorithms for adaptive decision making and development of decision trees.

A. Decision Tree Pruning

Decision tree pruning is a well established concept, that has been studied extensively over the years and there are various pruning techniques. But all of these techniques can be categorized into two groups. They are:

- **Pre-pruning:** In this strategy, the growth of decision trees are stopped early by enforcing different constraints on decision tree. For example, maximum depth of tree, minimum sample size to create new branch etc.
- **Post-pruning:** This approach first allows trees to grow fully and then remove branches with low contribution to predictive performance. One big advantage of this technique that it offer better generalized decision tree over pre-pruning approaches. This strategy includes Cost-Complexity Pruning (CCP) [9], Reduced-Error Pruning (REP) [10], Error-Based Pruning [11] etc.

B. Reinforcement Learning for Model Optimization

In ML, reinforcement learning has demonstrated advance optimization capability, particularly in the case of hyperparameter tuning, feature selection and neural architecture search. Traditional optimization techniques, such as grid search, Bayesian optimization etc., often struggle with high dimensional search space. As RL-based optimization technique learns from the interactions with the system and uses reward from actions to guide its search, it has proven itself as an alternative of the conventional techniques.

In hyperparameter optimization, [12] developed Hyp-RL, a policy based on Q-learning, which can navigate high dimensional hyperparameter spaces. Beyond the optics of hyperparameter optimization, RL has also been used for feature selection process, focusing the improvement of model interpretability as well as computational efficiency. [13] proposed a RL-based feature engineering approach, which finds the most informative features utilizing reward signals. RL has also shown promise in the optimizing network type machine

learning models, such as Neural Network, Decision Tree etc. [14] has proven the effectiveness of reinforcement learning in optimum neural architecture search operation. And for decision tree optimization, studies like [15] has demonstrated the performance of RL, leading to better generalization and predictive performance. These achievements highlights the potential of RL for optimizing ML models.

C. Multi-Armed Bandits for Efficient Model Selection

In optimizing sequential decision making, Multi-Armed Bandits provides an effective optimization mechanism. The most used MAB techniques, Upper Confidence Bound (UCB) algorithm studied in [16] and Thompson Sampling studied in [17], are widely adapted for balancing the exploration and exploitation in decision making. In ML, MAB has been utilized in various tasks. Like RL, it has been applied in feature selection purpose, hyperparameter tuning and optimizing neural network structure. [18] has proven MAB's applicability in identifying best set of features for training operation. While [19] and [20] display the usefulness of MAB algorithms in machine learning models' hyperparameter tuning. Additionally, [21] illustrated how MAB techniques can refine the neural network architecture by offering systematic pruning approach. All these studies emphasizes MAB's effectiveness in decision making tasks, aligning with the principles of reinforcement learning.

D. Multi-Armed Bandits Based Approaches for Decision Trees

With the increased popularity of decision tree algorithm, several studies have integrated MAB techniques into decision tree learning. The study [22], employs MAB algorithm to efficiently find split points during the construction of decision tree and increase the training speed. While [23] leverage MAB techniques to select the best set of attributes during decision tree induction with a goal to generate more cost-effective trees without compromising accuracy. However, these studies focus primarily on the growth of decision trees rather than pruning ineffective branches of those trees.

E. Summary and Research Gap

Although MAB-based strategies have demonstrated efficiency in model optimization, the study on their application to decision tree pruning still remains an unexplored area. Our work addresses this research gap by introducing a novel adaptive pruning mechanism for decision tree optimization based on Multi-Armed Bandits. It:

- Uses MAB-based decision making to identify branch nodes that offer relatively small performance to the model for pruning dynamically.
- Balances exploration and exploitation based on reward gained or lost from each action.
- Reduces the complexity of decision tree, making it more interpretable and optimized.

By leveraging the power of reinforcement learning, MAB-based pruning approach for decision trees could not only offer data-driven optimized decision trees but also improve the predictive performance.

III. METHODOLOGY

To implement MAB-based pruning of decision tree, first step is to define certain key aspects of MAB. As Multi-Armed Bandits is a simplified version of reinforcement learning, a single state reinforcement learning approach, defining action and reward is the most important part in this algorithm. Here, action means a branch node is being replaced with a leaf node in the decision tree, pruning the whole sub-tree from that branch node. To avoid underfitting, only branch nodes beyond level three are considered for pruning.

Additionally, the reward is being calculated using a function, utilizing model performance matrices like classification accuracy, logarithmic loss and f1-score. We split the dataset 65:35 into training and testing. Only 0.02% of the training data is used for consistent evaluation during pruning and for computing rewards, reducing computation compared to using the entire test set for MAB-based optimization. The test data is later being used to assess both pruned and unpruned models. The actions, selecting branch nodes to be pruned, along with the reward from them are being kept in history and these are later being used by MAB techniques to select future actions. This process is repeated for a fixed number of rounds. Figure 1 illustrates our proposed approach in a flowchart form.

The calculation of reward from pruning, a two step calculation, varies depending on which type of bandits techniques are being used. Although the first step of calculating reward in our proposed approach is similar for all techniques:

$$\Delta\text{Score} = \alpha(\text{Accuracy}_{\text{new}} - \text{Accuracy}_{\text{old}}) + \gamma(\text{F1}_{\text{new}} - \text{F1}_{\text{old}}) - \beta(\text{Loss}_{\text{new}} - \text{Loss}_{\text{old}}) \quad (1)$$

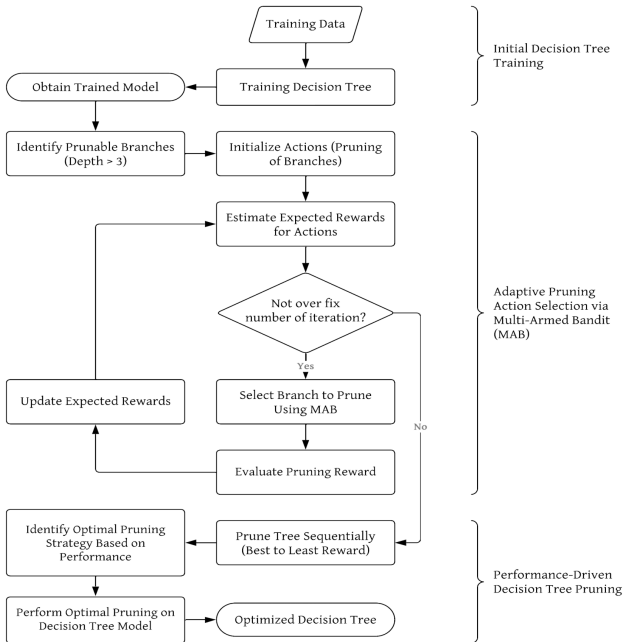


Fig. 1. Overview of the proposed decision tree pruning approach using multi-armed bandit for adaptive pruning and performance-driven optimization.

Here, $\alpha = 1$, $\beta = 1$, and $\gamma = 2.5$ are weight factors controlling the importance of accuracy, scaled logarithmic loss $[0, 1]$ and F1-score, respectively. These weight factor values have been kept consistent throughout our study. The delta score indicates the pruned tree's performance on a random training subset (0.02%) relative to the unpruned tree. However, the second step varies based on the chosen MAB technique.

For example, when we are using Upper Confidence Bound 1 (UCB1), Kullback-Leibler Upper Confidence Bound (KL-UCB) and Win-Stay, Lose-Shift (WSLS) techniques, the reward is being calculated using this function:

$$\text{Reward} = \frac{\max(0, \text{Threshold} + \Delta\text{Score})}{\text{Constant}} \quad (2)$$

where, the value of threshold determines how much loss in performance, negative value of ΔScore , is being tolerated after pruning. For example, if pruning results in a less performing tree with a delta value of -0.03, a threshold with a value of 0.05 can still result in a positive reward. Additionally, the value of constant is defined in such way that reward can be bounded between one and zero.

But for Thomson Sampling (TS) and Bayesian Upper Confidence Bounds (Bayes-UCB) techniques, we used a different approach as they follow Bernoulli rewards, which suggests:

$$\text{Reward} = \begin{cases} 1, & \text{if } \Delta\text{Score} > 0 \\ 0, & \text{if } \Delta\text{Score} \leq 0 \end{cases} \quad (3)$$

A. Algorithm

Given a decision tree model M , number of rounds T , delta function D and reward function R , we have divided the whole pruning operation into two steps. Each step has been displayed as algorithms. Algorithm 1 finds appropriate candidate branch for pruning and Algorithm 2 finds best nodes to prune to optimize decision tree.

Algorithm 1 Finding Prunable Decision Tree Branches

Require: Trained decision tree M

Ensure: List of prunable branches B

- 1: Initialize list $B \leftarrow \emptyset$
- 2: **for** each node b_i in M **do**
- 3: **if** b_i has no children **and** $\text{depth}(b_i) > 3$ **then**
- 4: Add b_i to B
- 5: **end if**
- 6: **end for**
- 7: **return** List B

In Algorithm 2, the section of performance evaluation with pruned tree, is dependent on which MAB technique is being used, as different MAB techniques use different approaches of reward calculation. But the calculation of score for each actions taken, later feed into the reward calculation, remains same for all cases.

Algorithm 2 MAB-Based Decision Tree Pruning

Require: Number of rounds T , reward function R , trained tree M , prunable branches $B = \{b_1, \dots, b_n\}$, hyperparameters α, β, γ

Ensure: Optimal pruned tree M_B

```
1: Initialize play count  $n_i \leftarrow 0$ , average reward  $\mu_i \leftarrow 0$  for each  $b_i \in B$ 
2: Evaluate unpruned tree  $M$ : compute accuracy  $A_M$ , loss  $L_M$ , F1-score  $f1_M$ 
3: Set best performance  $P_B \leftarrow -100$ , best model  $M_B \leftarrow M$ 
4: for  $t = 1$  to  $T$  do
5:   Select a random data subset  $S$ 
6:   if any  $b_i$  is untested then
7:     Select  $b_i$  randomly
8:   else
9:     Select  $b_i$  using MAB policy (e.g., UCB1, Thompson Sampling)
10:  end if
11:  Temporarily prune  $b_i$  to get  $M'$ 
12:  Evaluate  $M'$ : compute  $A_{M'}$ ,  $L_{M'}$ ,  $f1_{M'}$ 
13:  Compute performance change:
      
$$D_i = \alpha(A_{M'} - A_M) - \beta(L_{M'} - L_M) + \gamma(f1_{M'} - f1_M)$$

14:  Compute reward  $R_i = R(D_i)$ 
15:  Update estimates:
      
$$n_i \leftarrow n_i + 1, \quad \mu_i \leftarrow \frac{(n_i - 1)}{n_i} \mu_i + \frac{1}{n_i} R_i$$

16:  Restore  $M$ 
17: end for
18: for  $c = 1$  to  $|B|$  do
19:   Prune  $c$  lowest-ranked branches to obtain  $M_c$ 
20:   Evaluate  $M_c$ : compute  $A_{M_c}$ ,  $L_{M_c}$ ,  $f1_{M_c}$ 
21:   Compute  $P_c = \alpha A_{M_c} - \beta L_{M_c} + \gamma f1_{M_c}$ 
22:   if  $P_c \geq P_B$  then
23:      $M_B \leftarrow M_c$ ,  $P_B \leftarrow P_c$ 
24:   end if
25: end for
26: return Optimal pruned model  $M_B$ 
```

IV. EVALUATION

We evaluate our proposed MAB-based pruning approach using five benchmark classification dataset from UCI and Kaggle repositories. The datasets include Breast Cancer Detection [24], Pima Indian Diabetic Detection [25], Titanic Survival [26], Ionosphere [27] and Credit Card Fraud Detection [28]. During the evaluation process, our main focus was to evaluate the usability of the proposed approach against traditional methods used in this purpose.

A. Experimental Setup

To ensure consistency, a set of hyperparameters, optimized via GridSearchCV, was used to train all decision trees:

- **Max Depth = 7:** Balances model complexity and overfitting.

- **Min Samples per Leaf = 3:** Ensures reliable predictions at terminal nodes.
- **Min Samples per Split = 3:** Prevents overly complex splits.

The following conditions were uniformly applied for pruning decision trees using MAB algorithms throughout the study:

- **Min Pruning Level:** Nodes above level 3 were excluded to avoid underfitting.
- **Min Evaluation Samples:** 0.02% of training data
- **MAB Iterations:** Set to 1100 to balance exploration and exploitation.

As the MAB-based pruning technique relies on iterative evaluation of actions, the following performance metrics were used to assess the effectiveness of the pruned models:

- **Accuracy:** Measures the generalization capability of the pruned model on unseen data.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.

- **Logarithmic Loss (Log Loss):** Evaluates the confidence of probabilistic predictions, with lower values indicating better-calibrated models.

$$\text{Log Loss} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{i,j} \log(\hat{y}_{i,j})$$

where, N = number of samples, M = number of classes, $y_{i,j} = 1$ if sample i belongs to class j , otherwise 0, and $\hat{y}_{i,j}$ = predicted probability of sample i for class j .

- **F1-Score:** Evaluates ML-based models' predictive performance. It combines both precision and recall scores of any classification models.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

These three metrics were integrated into a single performance function to guide the MAB pruning process:

$$\text{Performance} = \alpha \cdot \text{Accuracy} - \beta \cdot \text{Log Loss} + \gamma \cdot F1$$

where $\alpha = 1$, $\beta = 1$, and $\gamma = 2.5$, as mentioned previously.

B. Empirical Evaluations of MAB-based Pruning Techniques

The primary goal of this experiment is to evaluate the impact of MAB-based pruning techniques on decision tree performance against traditional approaches, like CCP, REP etc. We only consider CCP for comparison as it is more sophisticated approach compared to REP and both offer similar performance in accuracy as discussed in [29].

For evaluation, we have applied the proposed technique on five different datasets and evaluated their predictive performance by utilizing the scoring function, described in Section

TABLE I
PERFORMANCE SCORES ACROSS DATASETS

Dataset	Unpruned	CCP	UCB1	TS	SM	WSLS	Bayes-UCB	KL-UCB	Mean Score
Diabetics	1.350	1.320	1.447	1.526	1.460	1.429	1.401	1.401	1.422
Credit Card	2.898	3.038	3.060	3.060	3.038	3.060	3.060	3.060	3.045
Breast Cancer	3.175	3.203	3.238	3.238	3.238	3.238	3.238	3.238	3.220
Ionosphere	2.854	2.948	3.038	3.038	3.038	3.133	3.038	3.038	3.021
Titanic	1.883	2.000	2.083	2.052	1.883	2.002	2.052	1.932	1.986
Mean Score	2.432	2.502	2.573	2.583	2.571	2.572	2.554	2.554	2.543

TABLE II
MEAN RANKS OF DIFFERENT METHODS

Method	Mean Rank
TS	2.8
UCB1	2.9
WSLS	3.1
Bayes-UCB	3.7
KL-UCB	4.4
Softmax	4.7
CCP	6.7
Unpruned Model	7.7

IV, on test data. Table I presents the performance scores for each pruning methods on the five datasets. Additionally, Table II presents the mean ranking of each method based on their overall performance.

To provide a formal comparison, we conduct an analysis on their statistical significance using paired t-tests, considering the unpruned decision tree and CCP-based pruned tree as baselines. Table III and Table IV provide a comparative analysis of our proposed method against both unpruned decision tree as well as the CCP-based pruned tree respectively.

1) *Performance Comparison Against the Unpruned Model:* The results in Table III demonstrate that all six MAB techniques used in the proposed approach outperforms the unpruned decision tree with performance based score improvement in between 4% and 6.5%. Among these, Thompson Sampling exhibits the highest score improvement with a value of 6.2%, which is followed by UCB1 and WSLS, both offering an improvement around 5.8%. To evaluate how these techniques differ in predictive classification statistically, we utilize paired t-tests, where a p-value < 0.05 indicates a significant difference between the pair of methods. Thompson Sampling exhibit the lowest p-value ($p = 0.0025$), suggesting a strong statistical significance, followed by UCB1 and Bayes-UCB with statistical significance, $p = 0.0058$ and $p = 0.0114$ respectively, depicting their effectiveness compared to unpruned decision tree. Although WSLS shows high level of performance improvement (5.77%), it's statistical significance ($p = 0.0221$) is slightly lower compared to Bayes-UCB ($p = 0.0114$), which offers lower performance improvement (5.17%). The rest show significant statistical significance, but their performance improvements vary, while Softmax method shows lowest performance improvement and least statistical significance 4.09% and 0.0351 respectively.

2) *Performance Comparison Against Cost-Complexity Pruning:* In this second evaluation, we perform the same tests, but this time we use the CCP-based pruned tree model as baseline model. The results from this evaluation has been summarized in Table IV. It shows different trend from the previous evaluation. Although TS still show highest

TABLE III
PERFORMANCE COMPARISON AGAINST UNPRUNED MODEL

Method	Mean Score	Improvement (%)	T-Statistic	P-Value
UCB1	2.5732	5.81%	5.3772	0.0058
TS	2.5828	6.20%	6.7770	0.0025
Softmax	2.5314	4.09%	3.1337	0.0351
WSLS	2.5724	5.77%	3.6315	0.0221
Bayes-UCB	2.5578	5.17%	4.4335	0.0114
KL-UCB	2.5338	4.19%	3.4658	0.0257

TABLE IV
PERFORMANCE COMPARISON AGAINST CCP METHOD

Method	Mean Score	Improvement (%)	T-Statistic	P-Value
UCB1	2.5732	2.85%	3.7278	0.0203
TS	2.5828	3.24%	2.4343	0.0716
Softmax	2.5314	1.18%	0.6770	0.5355
WSLS	2.5724	2.82%	2.0865	0.1052
Bayes-UCB	2.5578	2.24%	4.2988	0.0127
KL-UCB	2.5338	1.28%	1.1359	0.3195

performance score improvement, its statistical significance is not the most. Meanwhile, UCB1 provides a well-rounded performance with 2nd highest performance improvement along with the most statistical significance. Here again, the dynamics between WSLS and Bayes-UCB can be seen, as WSLS show higher performance improvement but lower statistical significance then Bayes-UCB. Among all , only UCB1 and Bayes-UCB methods have $p < 0.05$. Interestingly, Softmax remains the lowest-performing among the six MAB techniques.

Overall, based on empirical results, our proposed approach outperforms the baseline models, with UCB1 and Bayes-UCB yielding the most consistent and significant improvements, making them ideal for decision tree pruning.

V. LIMITATIONS

Although our Multi-Armed Bandits-based pruning approach shows promising results, several limitations should be noted. Firstly, the approach heavily depends on hyperparameter choices such as reward thresholds, accuracy, loss, F1-score weighting, and specific MAB algorithm selection, which might vary across datasets. Secondly, our experiments were limited to small and medium-sized datasets, and its scalability to larger, more complex real-world datasets remains uncertain. Additionally, despite using a smaller data subset for performance evaluation, our reinforcement learning-based approach still has higher computational overhead compared to traditional methods like CCP. Furthermore, we tested only classification tasks; the method's effectiveness on regression or other machine learning tasks has yet to be evaluated. Finally, the stochastic nature of MAB introduces variability, potentially requiring multiple trials to achieve consistent results.

In our future research, we should focus on extensive evaluations on diverse datasets, improved hyperparameter tuning, and extending the approach to other machine learning tasks.

VI. CONCLUSION

This study explores the utilization of MAB algorithms for pruning operation of decision tree, focused to not only reduce

the complexity of decision tree but also improve predictive accuracy. As decision tree models are quite prone to overfitting, developing an efficient way of generating optimal tree models is essential for better generalization of tree models. Results from our study, conducted on multiple benchmark datasets, indicate that MAB-based pruning techniques perform quite well, particularly UCB1 and Bayes-UCB, because of their all-round performance. As displayed in Table I, all MAB based pruning approaches discussed in our study, outperform the conventional pruning technique, cost-complexity pruning, while maintaining competitive performance across different datasets. Furthermore, the mean rank analysis result, illustrated in Table II, indicates the superior performance of MAB-based pruning techniques, suggesting their effectiveness in optimizing decision tree compared to traditional one. This idea is later proven in the t-test analysis, which is exhibited in Table IV.

Beyond predictive performance improvement, MAB-based pruning methods offer a probabilistic pruning operation rather than the fixed heuristic approach conventional method based on. This characteristic provide a better data-driven pruning decision which in turns improve the performance of decision tree.

Future research work can explore various extensions of our study. First, integrating contextual bandits into the pruning operation, allowing more adaptive and data-driven decision making for pruning based on contextual information like feature importance, dataset characteristics etc. Second, studying how MAB-based pruning approaches perform in ensemble methods might help to develop more robust and generalized tree-based machine learning models. Lastly, applying MAB-based pruning techniques to real-world application, particularly for medical diagnosis or financial risk assessment where interpretability is necessary, could provide insights about their effectiveness and scalability in complex real-world decision making tasks.

Developing an efficient pruning technique remains an open challenge, as a decision tree is regarded as a cornerstone of interpretable machine learning algorithms. In this study, we demonstrate a new way of adaptive pruning techniques for decision trees using MAB-based approaches and contribute to the growing research on advancement of interpretable machine learning models.

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